

Evidence of User Need

About this activity

Before diving into whether or not to use AI, your team should gather user research detailing the problem you're trying to solve. Use this to make a case for using AI to solve your user need:

1. Meet as a team, look at the existing user research and evidence you have, and detail the user need you're trying to solve
2. Next, write down a clear, focused statement of the user need
3. Then, read through each of our "AI suitability" statements to identify if your user need is a potential good fit for an AI solution
4. Finally, summarize your decision, clearly stating why or why not AI will help solve your user need

At the end of this exercise your team should be aligned on whether AI is a solution worth pursuing and why.

Before you begin:

- Block out time to get as many cross-functional leads as possible together in a room to work through these exercises together.
- The person responsible for user research should aggregate existing evidence for the team to reference in this and subsequent exercises. Complete and share the user research summary with cross-functional leads and activity stakeholders ahead of time.

Worksheet: User research summary

List out the existing evidence you have supporting your user need. Add more rows as needed. Share this summary with cross-functional leads and activity stakeholders ahead of time.

Date	Source	Summary of findings

1. Statement of User Need

Write down a clear, focused statement of the user need:

How might we solve _____**{ our user need }**_____?

How can AI solve this problem in a unique way?

When considering if and how AI might solve the problem in a unique way, consider:

How pervasive is the problem?	
Whom does it affect?	
How does this problem affect various individuals?	
How does this problem vary across different dimensions of identity or historically marginalized communities?	
Do the problem's effects change over time?	

2. AI suitability

Read through each of our "AI suitability" statements to identify if your user need is a potential good fit for an AI solution. You may add one or two more statements that describe your team's specific AI capabilities.

AI is probably better for	AI is probably **not** better for
☐ **Personalization:** The core experience requires personalizing content to individual users, or to a domain	☐ **Consistency:** The most valuable part of the core experience is its predictability, regardless of context or input
☐ **Predictions:** The core experience requires prediction of future events	☐ **Cost:** Running an AI solution at the needed frequency is very high
☐ **Scale:** Need to recognize a general class of things that is too large to articulate every case	☐ **Stakes:** The consequences of errors is very high and outweighs the benefits of a small increase in success rate
☐ **Volume:** Need to detect low occurrence events that are constantly evolving	☐ **Explainability:** Users, customers, or developers need to understand exactly everything that happens in the code and application
☐ **Language:** User experience requires natural language interactions, or agent or bot experience for a particular domain	☐ **Momentum:** Responsibly getting to market first is more important than other factors or the value using AI would provide
☐ **Diversification:** The user experience is significantly enhanced when outcomes are variable, rather than predictable	☐ **Directive:** People explicitly tell you they don't want a task automated or augmented
☐ {add your own}	☐ {add your own}

3. Summary of Evidence

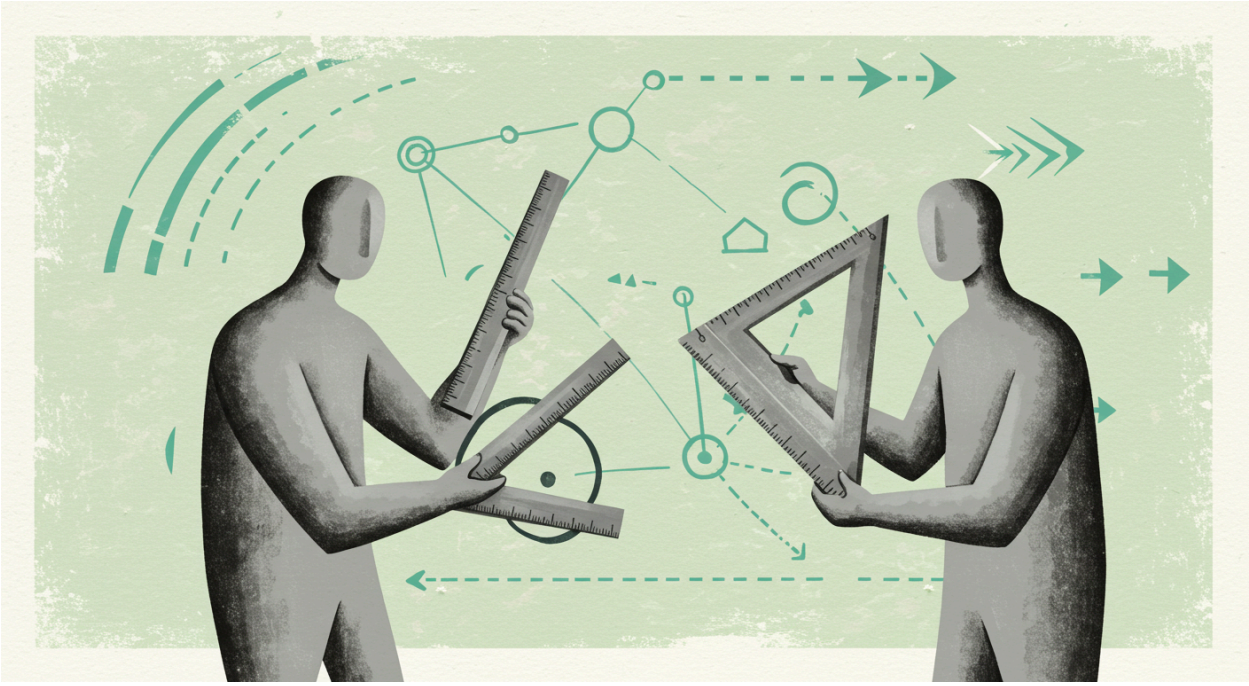
Summarize your decision, clearly stating why or why not AI will help solve your user need. AI capabilities and user contexts can change over time. This might require you to revisit your decisions, so it's important to be concrete about your rationale.

We think AI **{ can / can not }** help solve _____**{ user need }**_____,

because _____

_____

_____



Specify and align the AI Problem

About this activity

Specification alignment describes the ability to guide an AI system towards an intended outcome. For AI systems to be successful, it's important to define the purpose of the system. This can help you align the AI system's behaviors with your users' intentions:

1. Meet as a team, look at the existing user research you have, and identify product behaviors that are natural, useful, beneficial, and safe for users.
2. First, define the primary goal that your AI product/feature will solve for your users.
3. Then, identify the sub-goals that people must solve before or while addressing the larger problem.
4. Then, articulate parts of the problem that are frequently under-specified.

5. Finally, brainstorm some varied ways to optimize your problem.

Throughout this exercise, the engineering team should identify the model specifications to build AI systems that align with expected behaviors identified by the user experience team. At the end of this exercise your team should have a clear understanding of what constitutes an AI solution that is worth pursuing, and why.

Before you begin:

- Block out time to get as many cross-functional leads as possible together in a room to work through these exercises together.
- The person responsible for user research should aggregate existing evidence for the team to reference in this and subsequent exercises. Complete and share the user research summary with cross-functional leads and activity stakeholders ahead of time.

Worksheet: User research summary

List out the existing evidence you have supporting your user needs. Add more rows as needed. Share this summary with cross-functional leads and activity stakeholders ahead of time.

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1. Primary goal

Identify your user's primary goal. Then, translate this into specific tasks or outcomes that your AI system needs to successfully and reliably accomplish to help users meet their goal.

Primary Goal	
What problem should we solve for people with our product or application? What is the user's primary goal? What does the user want the AI system to do for them to help them meet their goals?	Our user's primary goal is **{ user goal }** _____ _____ To meet their goals, our AI system must **{ tasks to help users meet their goals }** _____ _____

2. Sub-goals

Consider the ecosystem of decisions and dependencies that surround your primary goal. Identify the sub-goals that people must solve before or while addressing the primary goal. Evaluate if and why these may be consistent or inconsistent with the primary goal.

Sub-goal	
What sub-goals must people solve before or while addressing the larger problem? What are the alternate goals, dependencies, or sub-problems that users break problems down into? What skills are required or what ancillary problems must users address before addressing the larger problem?	Sub-goals the user must address **before** addressing the primary goal: • **{ sub-goal }** _____ • **{ sub-goal }** _____ • **{ sub-goal }** _____ Sub-goals the user must address **while** addressing the primary goal: • **{ sub-goal }** _____ • **{ sub-goal }** _____ • **{ sub-goal }** _____
Are these consistent with the primary goal that the user wants solved?	These sub-goals are **{ consistent or inconsistent }** with the primary goal because _____ _____

3. Underspecification

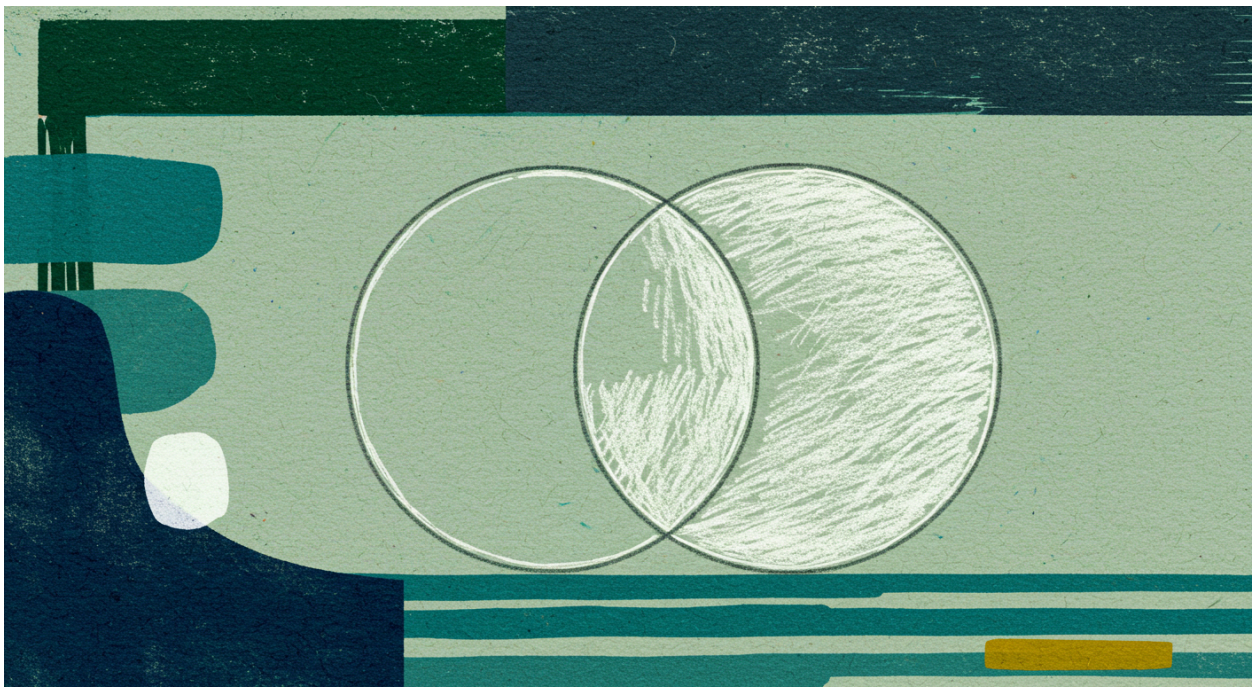
Articulate parts of the problem that are frequently under-specified, assumed, or simply overlooked, but may be critical to the solution. Repeat this as many times as necessary.

Underspecification	
What parts of the problem do people frequently under-specify? Consider frequently-made assumptions, tasks that are difficult to articulate or explain, and expectations and contextual cues that may not be visible to your AI product or feature.	If ___ **{ attribute of primary goal }** ___ _____ is underspecified or unknown, it can result in _____ **{ negative consequence }** _____ _____ _____
How will you uncover this knowledge? Consider frequently made assumptions, tasks that are difficult to articulate or explain, and expectations and contextual cues that may not be visible to your AI product or feature.	This information can be specified if we _____ **{ action or plan }** _____ _____ _____

4. Optimization

Misalignment between user intent and the AI system can cause an AI system to learn an unintended goal and competently pursue that goal in a new situation, also known as "reward hacking". Brainstorm some varied ways to optimize your problem. Consider if optimizing for one sub-goal can compromise another.

Optimization	
Can optimizing one sub-goal result in a compromise for others? What are some varied ways to optimize the problem? Can the AI system confuse the varied problems and learn harmful behaviors?	Meeting our primary goal can ... • **{ benefit to user }** _____ _____ • **{ benefit to user }** _____ _____ • **{ benefit to user }** _____ _____ But what if you AI model ... • **{ unexpected behavior }** _____ _____ • **{ unexpected behavior }** _____ _____ • **{ unexpected behavior }** _____ _____



Automation vs. Augmentation

About this activity

Automation occurs when a machine, software, or function performs a task with minimal to no user intervention. Augmentation occurs when a machine, software, or function amplifies the user's own capabilities to discover new insights and efficiencies collaboratively.

If your team has a hypothesis for why AI is a good fit for your user's need, conduct user research to further validate if AI is a good solution through the lens of automation or augmentation:

1. First, read through each of our considerations for automation & augmentation to identify ways in which automation and augmentation occurs in your domains.
2. Then, use contextual inquiries to understand opportunities for automation or augmentation.
3. Finally, create lightweight prototypes and evaluate them with potential users to understand initial reactions.

At the end of this activity, your team should have a clear picture of if and how AI should augment the user experience, which tasks should and shouldn't be automated, or what combination of automation and augmentation can create real value for your users.

1. Considerations for Automation & Augmentation

Read through each of our considerations for automation and augmentation statements. Based on these, identify when and how automation and augmentation might occur in the domain of your product.

Considerations for Automation	
☐ **Lack of ability or resources:** People need to perform repetitive tasks	When people need to perform a one-time task for which they lack expertise, or don't have enough time to manually execute, they may prefer to rely on automations that they can control at their discretion. E.g. tasks that are overly challenging, manual and urgent tasks, complex or multi-step tasks.
☐ **Boredom or repetition:** People have varying constraints when approaching a task	Sometimes people prefer to avoid tasks requiring substantial work in proportion to the value of results, especially when they need to perform the task frequently. E.g. Low value and tedious tasks.
☐ **Minimal oversight:** Tasks can be safely performed with little human guidance	Tasks that can be broken down into sequential steps, when followed exactly, have very little chance of error are usually better candidates for automating. When deciding which tasks can be automated, frame the task in terms of light or heavy oversight, and consider the consequences and cost of a bad prediction to a user. E.g. Low-risk tasks and predictable workflows.

Considerations for Augmentation	
☐ **Personal enjoyment:** People enjoy performing the task	When people enjoy a task, they often don't see it as a chore they want taken off their plate. However, augmentation with AI can have complicated effects on beginners and novices. E.g. Creative workflows
☐ **Personal responsibility** People are held accountable for consequences	Certain scenarios are associated with significant obligations and consequences for the individuals responsible. E.g. Decision-support and legally regulated workflows.
☐ **Significant or high risks:** People need to manage risks	People need to maintain control over AI-assisted decisions when risks are high or nuanced, especially those that promote safety, meet industry standards, and comply with local laws and directives. E.g. workflows and tasks with physical, emotional, socials and cultural risks.
☐ **Communicating preferences:** People have complex needs and preferences	People may not always express their intents or instructions accurately for AI systems to understand, such as when tasks are ambiguous or when using local language dialects or vernacular. E.g. Unstructured tasks and dialog-based systems
☐ **Complexity over time:** People engage with multi-step tasks over time	When it comes to intricate, lengthy, or collaborative workflows, people often prefer fine-grained control that lets them switch between tasks, use multiple products, and do things on their own time. Some of these can be inherently more complex than others for some people because of technological access or personal contexts. E.g. Collaborative or long-term user journeys

2. Contextual Inquiry

Contextual inquiry is a semi-structured interview method to learn about the contexts and ways in which users use a product or feature. During these interviews, you'll observe participants as they work, perform tasks, or use a product of interest in their own environments – or the natural context. You might ask your participants a set of questions that can help you understand what users are trying to do, what tasks they need to execute, how much they understand how an AI product works, and how much they know about what they're trying to do.

The analysis data produced by this method can be more realistic than laboratory data. Below are some example questions you can ask to learn about how your users think about automation and augmentation:

Research protocol: Example Questions

- If you were helping to train a new coworker for a similar role, what would be the most important tasks you would teach them first?
- Tell me more about that action you just took, is that an action you repeat:
    - Hourly
    - Daily
    - Weekly
    - Monthly
    - Quarterly
    - Annually
- If you had a human assistant to work with on this task, what, if any, duties would you give them to carry out?

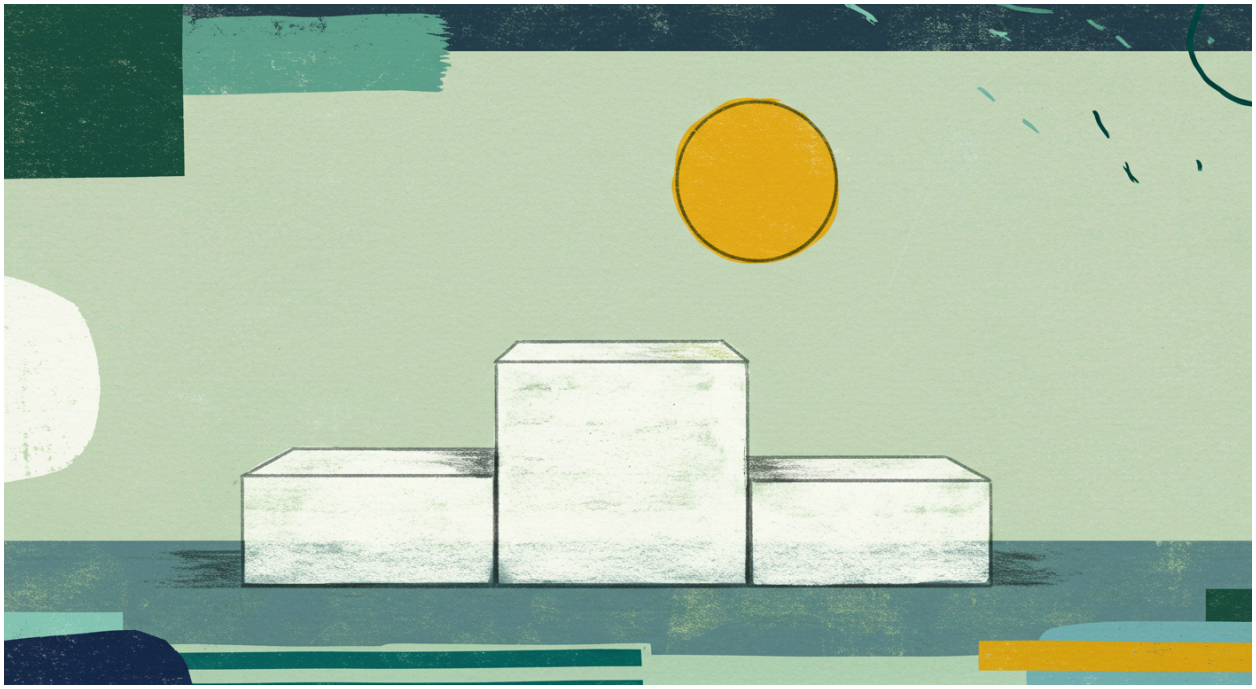
3. Test initial reactions

Based on your contextual inquiry, consider prototyping a selection of automation and augmentation solutions to understand initial user reactions. You might use this method as an alternative to contextual inquiries if going to meet your users in context isn't feasible.

The triptech method is an early concept evaluation method that can be used to outline user requirements based on likes, dislikes, expectations, and concerns.

Research protocol: Example Questions

- Describe your first impression of this feature.
- How often do you encounter the following problem: [insert problem/need statement here]?
    - Daily
    - Often (a few times a week)
    - Sometimes (a few times a month)
    - Rarely (a few times a year)
    - Never
- How important is it to address this need or problem?
    - Not at all important
    - Somewhat important
    - Moderately important
    - Very important
    - Extremely important



Design your reward function

About this activity

Once your team has had a chance to digest your recent research on user attitudes towards automation and augmentation, meet as a team to design your AI's reward function. You'll revisit this exercise as you continue to iterate on your feature and uncover new insights about how your AI performs.

1. First, use the confusion matrix template to list out examples of false positives and negatives, and true positives and negatives.

2. Then, evaluate the user benefit of false positives and negatives to decide your reward function or optimization strategy.

At the end of this exercise your team should have a clear understanding of what types of reward functions will most benefit your users.

1. Identify false positives & negatives

Use the template below to list out instances of when your AI system gets a true and false positive, and a true and false negative.

		Prediction	
		Positive	Negative
Reference	Positive	True positive _____ { Example 1 } _____ _____ { Example 2 } _____ _____ { Example 3 } _____	**False Negative** _____ { Example 1 } _____ _____ { Example 2 } _____ _____ { Example 3 } _____
	Negative	**False positive** _____ { Example 1 } _____ _____ { Example 2 } _____ _____ { Example 3 } _____	True negative _____ { Example 1 } _____ _____ { Example 2 } _____ _____ { Example 3 } _____

2. Plan your reward function or optimization strategy

Take a look at the false positives and false negatives your team has identified.

- If your feature offers the most user benefit for **fewer false positives**, consider optimizing for **precision**.
- If your feature offers the most user benefit for **fewer false negatives**, consider optimizing for **recall**.

Our AI model will be optimized for _____ **{ precision / recall }** _____ because _____ **{ user benefit }** _____ _____.

The tradeoff for choosing this method means our model will _____ **{ user impact }** _____ _____ _____.



Define Your Success Criteria

About this activity

AI-powered product experiences need to be updated on a regular basis to account for a variety of changes – such as bug fixes, the introduction of new features, or expanding to people in new geographic regions responsibly. It's important to add post-launch monitoring in your product development process to iteratively learn from user feedback and emerging harms, and to respond to these. One useful approach is to connect potential negative outcomes with measurable changes to the user experience to address them.

Once you've done the work to understand whether AI is a good fit for your user need and identified the tradeoffs of your AI's reward function, it's time to meet as a team to define success criteria for your feature. Through this exercise, your team may come up with multiple metrics for success.

1. First, use the success metrics framework to brainstorm success metrics.

2. Then, iterate on your success metrics against a basic checklist
3. Finally, create a plan to regularly meet and review your success metrics

By the end of this exercise, everyone on the team should be aligned on what success looks like for your AI feature, and how to alert the team if there is evidence that your AI feature is failing to meet the success criteria.

1. Success metrics framework

Start with this template and try a few different versions:

Template	If **_____ { specific success metric } _____** for **_____ { your team's specific AI driven feature }_____** **__ { drops below / goes above } __ { meaningful threshold } __** we will **_____ { take a specific action } _____** .
Version 1	
Version 2	

2. Statement Iteration

Take each version through this checklist:

- [ ] Is this metric meaningful for all of our users?
- [ ] How might this metric negatively impact some of our users?
- [ ] Is this what success means for our feature on day 1?
- [ ] What about day 1,000?
- [ ] How might this metric remain meaningful over time?
- [ ] How might this metric shift when AI adapts to users?
- [ ] How will you know that this metric is robust to changes in user intent and inputs?
- [ ] Add your own

Final Version	

3. Schedule regular reviews

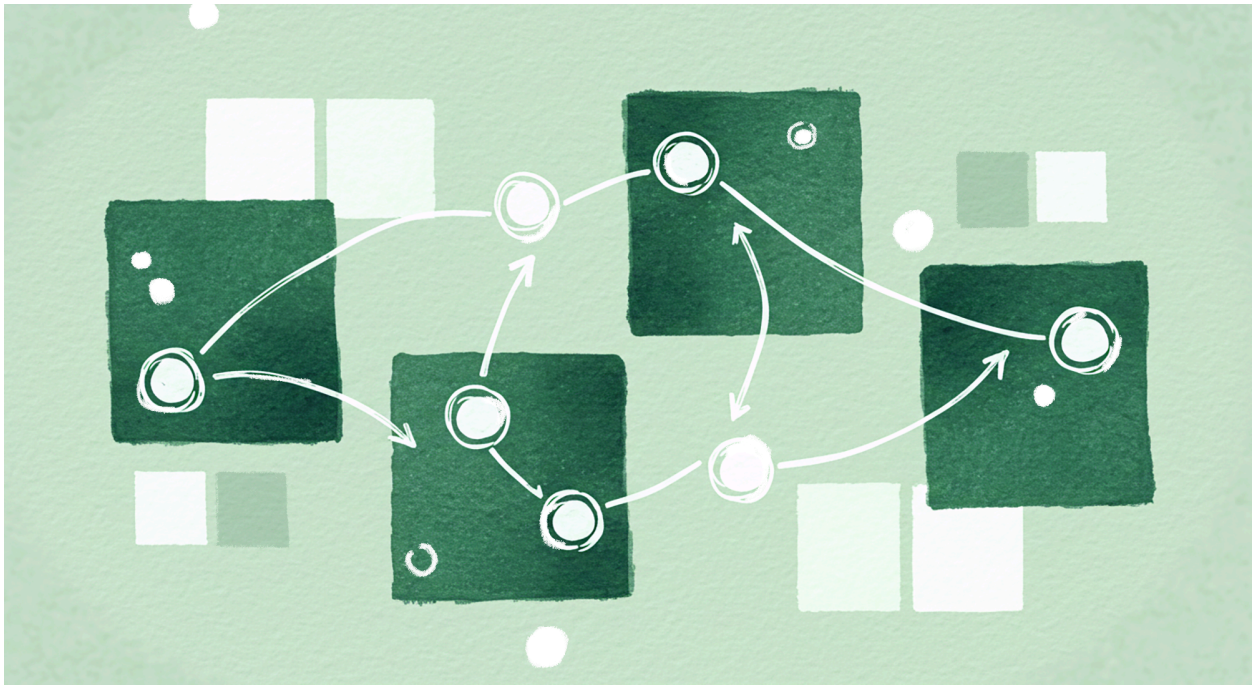
Once you've agreed upon your success metric(s), put time on the calendar to hold your team accountable to regularly evaluate whether your feature is progressing towards and meeting your defined criteria.

Success metric review

Date:

Attendees:

Agenda:



Create Interaction Design Policies

About this activity

Interaction design policies are the set of criteria that govern the user's experience at the point in a critical user journey (CUJ) where they interact with an AI system in your product. These describe acceptable and unacceptable tasks that an AI system can perform when the user interacts with it, potential risks from errors, and levels of uncertainty that the users can safely engage with. Create interaction design policies to define how the product responds to different inputs from people at critical moments, to ensure a safe and beneficial product experience:

1. First, identify the most important critical moments in your user journeys.
2. Then, as a team, brainstorm and define all four criteria of your interaction design policy for each critical moment.

3. Finally, translate each criteria into UX and ML requirements to work towards.

By the end of this exercise, everyone on the team should be aligned on what the safe and beneficial interactions with AI are for your product and how you'll enable these. Stakeholders should walk away with actionable interventions, such as user journeys and interactions in the user experience, and requirements for the AI systems in your product.

1. Identify critical moments

Identify critical moments within the product experience when users interact with an AI system. You'll craft your interaction design policy around these. There can be multiple critical moments in your product's CUJ, so prioritize critical moments that can significantly change the course of the user journey, or where AI outcomes can be more uncertain.

Use the table below to identify critical moments that need interaction design policies in your product. Add rows as necessary:

Critical User Journey	Critical Moment
Write your product's CUJs :	Prioritize critical moments in the CUJ :
As a **_____ { user } _____** I want to **_____ { goal } _____** So I **_____ { task # 1 } _____** **_____ { task # 2 } _____** **_____ { task # 3 } _____**	• **_____ { critical moment } _____** • **_____ { critical moment } _____** • **_____ { critical moment } _____**
As a **_____ { user } _____** I want to **_____ { goal } _____** So I **_____ { task # 1 } _____** **_____ { task # 2 } _____** **_____ { task # 3 } _____**	• **_____ { critical moment } _____** • **_____ { critical moment } _____** • **_____ { critical moment } _____**

2. Define your Interaction Design Policies

Interaction design policies are made up of four key sets of criteria—acceptable actions, unacceptable actions, thresholds of uncertainty, and vulnerabilities. Use the prompts provided to brainstorm and define each of these for each of your critical moments:

Acceptable Actions	
What are the actions, tasks, or behaviors that your product's AI should perform well for AI to add value to the user's experience? "We want {people or users} to be able to use our product's AI to / for {goal-based tasks and behaviors to encourage}." "Our product's AI should always {input/output constraints, measurable, behavioral criteria}."	• **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____**

Unacceptable Actions	
What are actions or tasks, and behaviors that your product's AI shouldn't perform, even if the user unintentionally asks for these? "We don't want {people or users} to intentionally or intentionally use our product's AI to / for {tasks and behaviors to discourage}." "Our product's AI should not {artifacts or aspects that the model should not produce}."	• **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____** • **_____ { criteria } _____**

Thresholds Of Uncertainty

What are the necessary criteria that AI outcomes need to meet to be shown to the user?

"If our AI's predictions are weak, {people or users} won't mind {actions, tasks, or workarounds} as long as {conditions & prerequisites}."

"{People or users} should be able to use a partially correct AI prediction for / if {tasks, conditions, or decisions}."

"To complete a partially performed AI task, {people or users} should be able to {tasks, features, or workarounds}."

- **_____ { criteria } _____**
- **_____ { criteria } _____**
- **_____ { criteria } _____**

Vulnerabilities

What are the different types of AI errors and subsequent risks users are vulnerable to?

"An incorrect or wrong generative AI prediction can harm {individuals and groups} when / if / by {actions, situations, or contexts.}."

"{People or users} may find themselves vulnerable if the AI outcomes result in {actions, situations, or contexts}."

- **_____ { criteria } _____**
- **_____ { criteria } _____**
- **_____ { criteria } _____**

3. Define UX and ML requirements

Have a rich discussion with cross-functional stakeholders and translate the interaction design policies into actionable steps. This may look like changes to product flows or interactions in the user experience, or to the requirements of the AI systems in your product. Other steps that teams can take may be privacy and security interventions, additional AI evaluations, or new data collection efforts. Where possible, use concrete examples grounded in user research and expected model behaviors.

_____ { your critical moment } _____			
	Criteria	UX requirements	ML requirements
Acceptable Actions			
Unacceptable Actions			
Thresholds Of Uncertainty			
Vulnerabilities			

Additional:

Resources:

- PRD
- Design Doc
- UX Research
- etc.

Team notes:

-

Next steps:

- [ ]

Parking lot:

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